

Infinite Basic Laver Patterns (IBLP): a formal specification

Definition 1 (IBLP). *An infinite basic Laver pattern (IBLP) is a quadruple*

$$A = (n, (A_1, \dots, A_n), (l_1, \dots, l_n), (M_1, \dots, M_n)),$$

where:

- (1) $n \in \mathbb{N}$ (allowing $n = 0$).
- (2) For each $i \in \{1, \dots, n\}$, A_i is a finite strictly increasing sequence of nonnegative integers such that $\text{len}(A_i) \geq 2$ and $\max(A_i) = i$.
- (3) For each $i \in \{1, \dots, n\}$, $l_i \in \mathbb{N}$ is the step length of row i .
- (4) For each $i \in \{1, \dots, n\}$, $M_i \subseteq A_i$ is the set of marked elements of row i .

All indices start at 1. We write $A_i[k]$ for the k -th element, and $A_i[-j] := A_i[\text{len}(A_i) - j + 1]$ for the j -th element from the end.

Definition 2 (short / medium / long). Row i is long if $\text{len}(A_i) > 2l_i$, medium if $\text{len}(A_i) = 2l_i$, and short if $\text{len}(A_i) < 2l_i$.

Definition 3 (zero / successor / limit). If $n = 0$, A is the zero pattern. If $n > 0$ and $\text{len}(A_n) = 2$ and $\min(A_n) = 0$, then A is a successor pattern. Otherwise A is a limit pattern.

Definition 4 (row comparison key). Let $A_i = (a_1 < a_2 < \dots < a_m)$ and $L = l_i$. Define the kept subsequence

$$K_i := \begin{cases} (a_1, a_2, \dots, a_m), & L \leq 1; \\ (a_1, a_{L+1}, a_{L+2}, \dots, a_m), & L \geq 2 \text{ and } m \geq L; \\ (a_1, a_2, \dots, a_m), & L \geq 2 \text{ and } m < L. \end{cases}$$

The row key is $\overleftarrow{K_i}$ (i.e. K_i reversed, compared from large to small).

Definition 5 (pattern order). Given two patterns A, B , compare their row keys $\overleftarrow{K_1(A)}, \overleftarrow{K_2(A)}, \dots$ and $\overleftarrow{K_1(B)}, \overleftarrow{K_2(B)}, \dots$ row by row in lexicographic order; the first difference decides. If all rows up to $\min(n_A, n_B)$ agree, the one with fewer rows is smaller.

Definition 6 ($p(i)$). For $i \in \{1, \dots, n\}$ define

$$p(i) := A_i[-(l_i + 1)].$$

Definition 7 (transfer sequence $tr_i(j)$). Fix row i . If $j = A_i[k]$ with $1 \leq k \leq \text{len}(A_i)$ and $k > l_i$, define the threshold $\theta := A_i[k - l_i]$. Let $t_0 := j$ and, whenever $t_m > \theta$ and $p(t_m)$ is defined, set $t_{m+1} := p(t_m)$. If there exists a least $m \geq 0$ with $t_m = \theta$, define

$$tr_i(j) := (t_0, t_1, \dots, t_m);$$

otherwise $tr_i(j)$ is undefined. If $k \leq l_i$, it is also undefined.

Definition 8 (partial function f_A). Assume A is nonzero and has $n \geq 1$ rows. Let $a_1 := A_n[1] = \min(A_n)$, $L := l_n$, and $p := p(n) = A_n[-(L + 1)]$. Define the partial function $f_A : \mathbb{N} \rightarrow \mathbb{N}$:

- (1) If $x < a_1$, then $f_A(x) = x$.
- (2) If $x = A_n[k]$ and $x < p$ and $k + L \leq \text{len}(A_n)$, then $f_A(x) = A_n[k + L]$.
- (3) If $x \geq p$, then $f_A(x) = x + (n - p)$.
- (4) Otherwise $f_A(x)$ is undefined.

Definition 9 (rows and step lengths of copy). Let A have $n \geq 1$ rows and set $p := p(n)$. If any required value $f_A(x)$ is undefined during copying, then $\text{copy}(A)$ is undefined. Otherwise define $A' = \text{copy}(A)$ to have $n' := 2n - p$ rows, satisfying:

- (1) For $1 \leq i \leq n - 1$: $A'_i = A_i$, $l'_i = l_i$, $M'_i = M_i$.
- (2) For each $i \in \{1, \dots, n - p + 1\}$, let the source row be $\sigma := p + i - 1$ and the destination row be $\tau := n - 1 + i$, and define

$$A'_\tau := \text{sort}(\{f_A(x) \mid x \in A_\sigma\}), \quad l'_\tau := l_\sigma,$$

requiring that $f_A(x)$ is defined for all $x \in A_\sigma$.

Definition 10 (mark filtering in copy). Continue the previous definition. For $x \in A'_\tau$, let $y \in A_\sigma$ be its unique preimage under f_A (so $f_A(y) = x$). Then $x \in M'_\tau$ iff:

- (1) $y \in M_\sigma$, and one of the following holds:

- (a) $tr_\sigma(y)$ is defined and $tr_\sigma(y)[-2] \geq p$; or
- (b) Let y_0 be the first element of $tr_\sigma(y)$ that is $< p$ (equivalently the largest element below p , since tr decreases). Either $y_0 < a_1$; or there exists k such that $y_0 = A_n[k]$ and $k + L \leq \text{len}(A_n)$ and $A_n[k + L] \in M_n$, and moreover if x has position q in A'_τ (1-indexed), then whenever $q + 1 - l_\sigma \geq 1$ we require

$$A'_\tau[q + 1 - l_\sigma] \leq a_1.$$

Definition 11 ($E(A, 0)$). If A is nonzero with $n \geq 1$, then $E(A, 0)$ is obtained by deleting the last row (row count becomes $n - 1$), leaving all earlier rows, step lengths, and marks unchanged.

Definition 12 (copy stage of $E(A, m)$ for $m \geq 1$). Assume A is neither the zero pattern nor a successor pattern. Let $A^{(0)} := A$. If the first copy($A^{(0)}$) is undefined, declare A a bad pattern and set $E(A, m) := E(A, 0)$. Otherwise define

$$A^{(1)} := \text{copy}(A^{(0)}), A^{(2)} := \text{copy}(A^{(1)}), \dots, A^{(m)} := \text{copy}(A^{(m-1)}),$$

and let

$$B := E(A^{(m)}, 0).$$

Now perform completion on B , starting at row index $r := n$ (where n is the row count of the original A). A temporary record map Rec (row index $\mapsto$ finite sequence) may be used and is discarded at the end.

Definition 13 (mark completion). Fix the current row r . Let

$$\mathcal{B} := \{ b \in M_r \mid b \in A_r \},$$

and enumerate $b_1 < \dots < b_k$ (this set is frozen at the start of this pass; newly created marks are not added). For each b_i in order:

- (1) If $tr_r(b_i)$ is undefined, skip.
- (2) Let $t := tr_r(b_i)[-2]$. If $\text{Rec}(t)$ is missing or empty, skip; otherwise set $s := \text{Rec}(t)$.
- (3) Check for all $j = 3, 4, \dots, \text{len}(tr_r(b_i))$ that

$$tr_r(b_i)[-j + 1] + 1 \in A_{tr_r(b_i)[-j]}.$$

If the check fails, skip.

- (4) If it passes, insert s and $b_i + 1, \dots, b_i + \text{len}(s)$ into A_r , then re-sort and deduplicate so the row remains strictly increasing. Update marks: elements of s are unmarked; $b_i + 1, \dots, b_i + \text{len}(s)$ are marked; other marks are preserved (subject to these rules). Update the step length: $l_r := l_r + \text{len}(s)$.

Definition 14 (native completion). Fix the current row r . If $\text{len}(A_r) > 2l_r$ (a long row), do nothing and return 0. Otherwise let

$$p_r := p(r) = A_r[-(l_r + 1)], \quad a := A_r[-l_r].$$

Construct a sequence s as follows: set $u_0 := a$ and, while $A_{u_m}[-2]$ exists and is $> p_r$, set $u_{m+1} := A_{u_m}[-2]$ and append u_{m+1} to s . Let $t := \text{len}(s)$. If $t = 0$ return 0; if $t > 0$ set $\text{Rec}(r) := s$ and proceed:

- (1) Insert t new rows immediately after row r , so all original rows with index $> r$ have their row indices increased by t .
- (2) Only for those original rows with index $> r$, shift within that row every element $> r$ by adding t ; shift marks accordingly; keep step lengths unchanged.
- (3) Let the old row and step length be $A_r^{\text{old}}, l_r^{\text{old}}$. Replace it by $t + 1$ rows:

$$A_{r+t} := \text{sort}(A_r^{\text{old}} \cup s \cup \{r + 1, \dots, r + t\}), \quad l_{r+t} := l_r^{\text{old}} + t.$$

Marks on row $r + t$: old marks are kept; $r, r + 1, \dots, r + t - 1$ are marked; s is unmarked; $r + t$ is unmarked.

- (4) For $i = t, t - 1, \dots, 1$, build A_{r+i-1} from A_{r+i} by deleting the element $r + i$, deleting the element $A_{r+i}[-l_{r+i}]$, removing the mark on $r + i - 1$, and setting $l_{r+i-1} := l_{r+i} - 1$.
- (5) **Medium-row exception:** if $i = t$ and A_r^{old} is medium (i.e. $\text{len}(A_r^{\text{old}}) = 2l_r^{\text{old}}$), then when constructing A_{r+t-1} do not delete $A_{r+t}[-l_{r+t}]$ and do not decrement the step length (set $l_{r+t-1} := l_{r+t}$), but still delete $r + t$ and still remove the mark on $r + t - 1$.

Native completion returns t .

Definition 15 (completion loop for $E(A, m)$). Initialize $r := n$ (where n is the row count of the original A). While r does not exceed the current number of rows of B , first perform mark completion on row r , then native completion on row r obtaining t , and update $r := r + t + 1$. When r exceeds the row count, completion stops, Rec is discarded, and the resulting B is $E(A, m)$.